ModelIt!

Student Activity Pack

Why Are You Taller Than Your Grandparents?

Modeling Gene Expression, Nutrition, and the Secular Trend in Human Height

Grade 10L1 | NGSS HS-LS3-2, HS-LS3-3 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Are you taller or shorter than your parents were at your age? What about your grandparents?

2 How much of your height do you think is determined by your genes versus your environment?

3 Draw a diagram showing what factors you think contribute to how tall you grow.

4 Why do you think the average height of people in many countries has increased over the past century?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Gene Expression Level | Growth Hormone Release | Nutrition Quality | Bone Growth Rate | Final Height

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Optimal Nutrition

Nutrition Quality: Maximum | All other factors responding

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Nutritional Deprivation

Nutrition Quality: Low | Simulating historical conditions

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Generational Comparison

Same genes | 1940s vs. 2020s nutrition

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Sleep Quality — The duration and depth of sleep, particularly deep sleep stages when growth hormone is released in its largest pulses — chronic sleep deprivation during childhood can significantly reduce growth

Physical Activity Level — Exercise stimulates growth hormone release and bone remodeling — weight-bearing exercise increases bone density, and moderate exercise promotes growth plate activity

Stress Hormone Levels — Chronic stress increases cortisol, which inhibits growth hormone release and can suppress growth plate activity — childhood adversity has measurable effects on growth trajectories

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If your DNA is inherited from your grandparents, how can you be significantly taller than they were at your age? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that identical genes can produce different heights depending on environment?**

2. **What does the 80/20 genetics-environment split mean for predicting height — can we predict adult height from DNA alone?**

3. **Why is the secular trend in height evidence for environmental influence rather than genetic evolution?**

4. **How does the concept of epigenetics challenge the traditional nature-versus-nurture debate?**

5. **What are the limitations of modeling a complex biological process like growth with only five components?**

8 STEM Challenge: Design a Childhood Nutrition Optimization Program

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YOUR ENGINEERING MISSION

Design an evidence-based nutrition program for a school district that maximizes growth potential and health outcomes for students, particularly those from food-insecure backgrounds.

THE SCENARIO

A school district serving 15,000 students has identified that 40% of its students are food insecure and many are below expected height-for-age percentiles. The superintendent has hired your team to design a comprehensive school nutrition program that uses the science of gene expression and growth to optimize student health outcomes within a realistic budget.

Engineering Design Thinking

1. WHAT WILL A GROWTH-OPTIMIZED SCHOOL BREAKFAST AND LUNCH LOOK LIKE BASED ON THE SCIENCE OF BONE GROWTH?
2. HOW WILL YOU ENSURE STUDENTS FROM FOOD-INSECURE HOMES RECEIVE ADEQUATE NUTRITION OUTSIDE SCHOOL HOURS?
3. WHAT ROLE SHOULD NUTRITION EDUCATION PLAY ALONGSIDE DIRECT FOOD PROVISION?
4. HOW WOULD YOU MEASURE THE PROGRAM'S IMPACT ON STUDENT GROWTH TRAJECTORIES OVER 3-5 YEARS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Nutritional Scientists study how food components affect gene expression, growth, and health. They work in research institutions, public health agencies, and food companies, earning $55,000–$110,000/year. Endocrinologists who specialize in growth hormones and developmental biology earn $200,000–$350,000/year as medical specialists.